Telecom Network Operations Assistant - API Documentation

Service topology

Synapt MCP bridge: http://127.0.0.1:8000/mcp.

Starlette backend API: http://127.0.0.1:8001.

React/Vite frontend: http://127.0.0.1:8790.

GET /api/health

Purpose: check backend liveness and whether the orchestrator has initialized.

Response: { status: ’ok’, orchestrator_initialized: boolean }.

POST /api/analyze

Purpose: run guardrail, query planning, Synapt context, incident retrieval, prediction or historical baseline, RCA, classification, impact, recommendations, summary, and response assembly.

Request fields: query string required; network_metrics object optional; prediction_mode is network_performance or fault_tolerance; top_k is an integer from 1 to 50.

Prediction behavior: live models run only with required telemetry; without telemetry, evaluation-style queries can produce a historical baseline.

Response: structured JSON containing final_answer, user_answer, sections, trace, execution metadata, and optional evaluation output.

POST /api/predict

Purpose: run the prediction agent directly without the complete assistant workflow.

Request: network_metrics object required and prediction_mode optional.

Response: prediction status, predicted throughput/fault risk, model trace, missing-feature information, and model limitations.

POST /api/classify

Purpose: classify and prioritize a supplied current metric scenario.

Request: network_metrics object required.

Response: configured dataset-relative risk labels and classification details; these are not telecom SLAs.

POST /api/feedback

Purpose: save user feedback for a response.

Request: response_id, query, final_response, rating (helpful or not_helpful), and optional comment/context.

Response: accepted status, feedback_id, response_id, and whether it was marked for evaluation.

Example /api/analyze request

{ "query": "Calculate historical averages for high latency and packet loss, assess network impact, and suggest troubleshooting actions from Synapt SOPs.", "prediction_mode": "network_performance", "top_k": 15 }

Error behavior

Invalid JSON or validation errors return HTTP 400/422 with a detail field.

Prediction model execution errors return HTTP 500 with a safe error message and diagnostic field.

The API is configured for local frontend origins on port 8790 and should remain bound to localhost during development.